Individual Assignment

Instructions

- Although students are placed in groups, **each student must answer the assigned questions individually**.
- Students within the same group will work on the **same questions**, but **each student must prepare and submit their own independent answers**.
- Any form of **copying or similarity between submissions will be considered plagiarism**.
- If two or more students are found submitting similar or identical work, **all involved students will receive a score of zero (0)**.
- Make sure your work reflects your **own understanding, explanation, and examples**.
- The individual assignment must be submitted on the same day as your group project presentation.

Group 1

1. Explain the concept of **Java I/O classes**. Describe different types of streams and how they are used in Java.
2. What is **Multithreading in Java**? Explain its importance and how threads are created and managed.

Group 2

1. Discuss **Object I/O in Java**. Explain serialization and deserialization with examples.
2. Explain the concept of **Networking in Java** and describe how client-server communication works.

Group 3

1. Explain **File management in Java**. Describe how to create, delete, and manipulate files using Java.
2. What is **Remote Method Invocation (RMI)**? Explain its architecture and advantages.

Group 4

1. Discuss **File and File Dialog objects** in Java. Explain their usage with examples.
2. Explain **Servlets in Java**. Describe their lifecycle and importance in web applications.

Group 5

1. What are **Random Access Files**? Explain how they differ from sequential file access.
2. Explain different **thread life cycle stages** in Java with a diagram.

Group 6

1. Describe the **basic structure and working of Java Streams** with examples.
2. Explain **Socket programming in Java** and how it is used in networking.

Group 7

1. Explain how **file handling is performed in Java** with appropriate examples.
2. Compare **Multithreading and Single-threading**. Discuss their advantages and disadvantages.

Group 8

1. Discuss **byte streams and character streams** in Java with examples.
2. Explain how **RMI is implemented in Java** step by step.

Group 9

1. Explain **exception handling in file operations** and why it is important.
2. Describe how **Servlets interact with web servers** and handle client requests.

Group 10

1. Discuss the importance of **File I/O in real-world applications** with examples.
2. Explain the advantages and challenges of using **Multithreading and Networking together** in Java applications.